

The Failure of DO_2 – Did What I Do to the Patient Just Work?

By now, I wouldn't be surprised if you were asking yourself, "I thought this was a book about ECMO?" Why are we talking about hemoglobin dissociation, shock recognition, shunt physiology, and lactate clearance?

Hopefully this chapter will put it all together.

We have been building toward a fundamental concept that we will unpack more in this chapter, namely, that everything we do in the intensive care unit (ICU) comes with a cost. There is toxicity associated with interventions, and this toxicity is often dose related.

Let me underscore that one more time, to hopefully make sure we appreciate its implications.

Every intervention in the ICU has a dose-related toxicity – the more we have to dial up support, the more we have to consider the potentially toxic effects of what we are doing (Fig. 5.1).

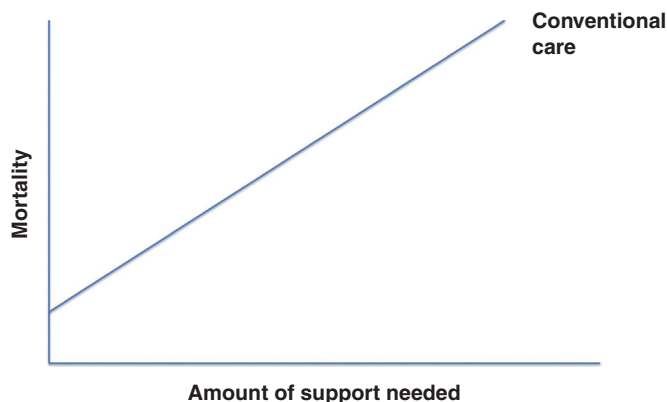


FIG. 5.1 The dose-related toxicity of conventional care

Let's return to the oxygen delivery equation:

$$\text{DO}_2 \rightarrow \text{CO} \times \text{Hb} \times \text{SaO}_2$$

We are going to unpack the concept of dose-related toxicity for each component, highlighting the cost of optimizing each component, especially as things start to break down as patients begin to decompensate.

WHAT IS MEANT BY DOSE-RELATED TOXICITY OF ICU INTERVENTIONS?

Let's say you have a patient who is septic from a urinary tract infection, with *Escherichia coli*, a gram-negative bacteria, growing in the urine as well as blood cultures. He is placed on vancomycin,

an antibiotic with gram-positive coverage but poor gram-negative coverage. When he does not improve, the dose of vancomycin is increased 20-fold.

What is wrong with this story? Why do we all cringe at this plan?

There are two fundamental flaws to this plan:

1. You have to know what you are treating
2. If you are treating the wrong diagnosis, increasing the dose will only expose this patient to toxicity without any benefit

IS THIS REALLY A FAIR EXAMPLE?

Let's entertain another example. In this case, it is a patient who arrives in the ICU with shock due to a pulmonary embolism with primarily right ventricular failure. The team assesses the patient, sees a low blood pressure, high heart rate, and low urine output and gives fluid.

When this doesn't work, they give another bolus of fluid. And then a third.

After this doesn't work, they initiate norepinephrine, with no improvement in blood pressure, uptitrating from 0.1 mcg/kg/min to the maximum of 2 mcg/kg/min. After this doesn't work, the team initiates vasopressin, uptitrated to a maximum rate of 0.04 units/min. After this, with the patient still hypotensive, the team initiates a neosynephrine drip, uptitrating it to a maximum of 200 mcg/kg/min.

WHAT IS HAPPENING WITH THIS PATIENT?

Much like our first patient who was being treated with escalating doses of the wrong antibiotic, this patient was exposed to escalating doses of support that was not helping his underlying condition. Let's explore why.

In a pulmonary embolism, there is an occlusive thrombus in the pulmonary artery, which can increase right ventricular afterload and lead to right heart failure.

Recall from [Chapter 2](#) that preload has a limited effect on a failing right ventricle. This is due to the various characteristics of the right ventricle – it is less muscular, shortens on a different axis, and this is eventually limited by the pericardium ([Fig. 5.2](#)).

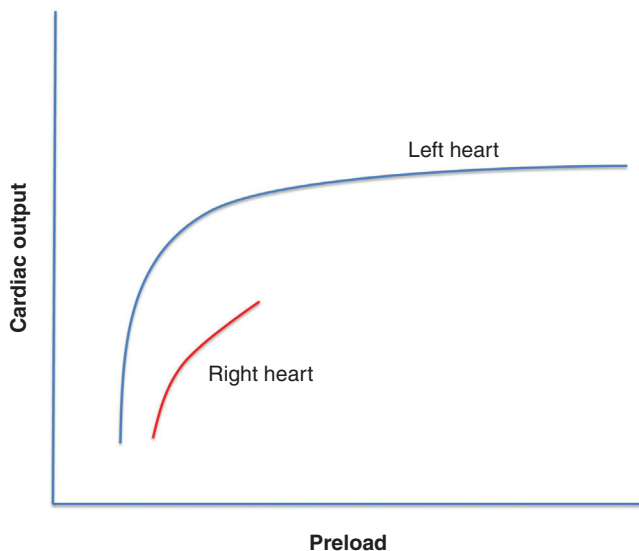


FIG. 5.2 Preload responsiveness for the left and right heart

Additionally, recall that the failing right ventricle will not only dilate but can actually compress on the left ventricle, as is illustrated here. This can be the unfortunate consequence of excessive fluid bolus in right heart failure (Fig. 5.3).

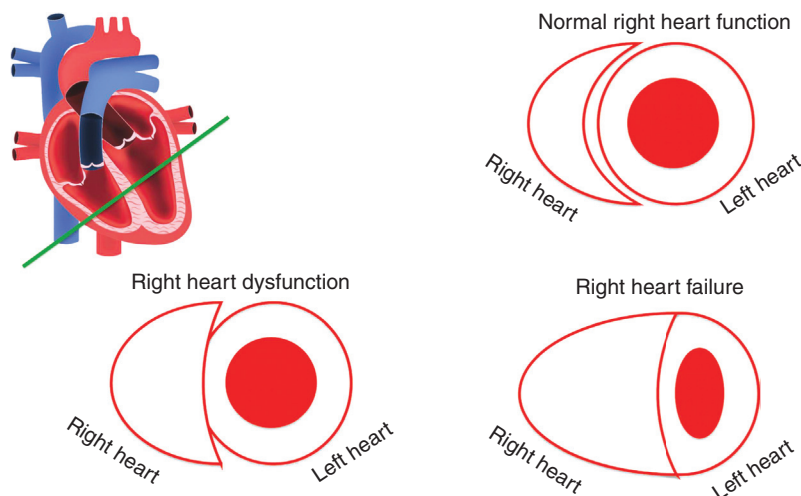


FIG. 5.3 The effect of excessive preload on a failing right ventricle. (Modified from [Ody_Stocker/Shutterstock.com](#).)

Next, the team initiates norepinephrine, which has the effect of improving systemic vascular resistance by increasing vasoconstriction, but remember also increases the pulmonary vascular resistance, which will only worsen right ventricular afterload (Fig. 5.4).

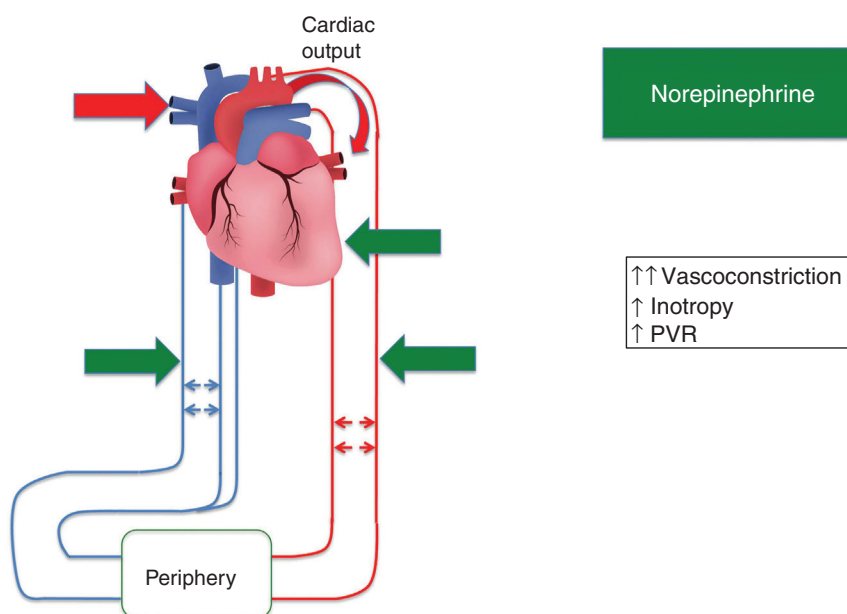


FIG. 5.4 Effect of norepinephrine on pulmonary vascular resistance. *PVR*, Pulmonary vascular resistance. (Modified from [Ody_Stocker/Shutterstock.com](#).)

Uptitrating norepinephrine and neosynephrine was the equivalent of increasing vancomycin to 20g – uptitrating a medication that is not improving the clinical situation while only causing more adverse effects.

HOW DO PRESSORS WORSEN THE CLINICAL SITUATION IN SHOCK?

One of the most important clinical effects of a mean arterial pressure (MAP) is maintaining a vascular pressure head to allow forward flow of blood and perfusion of peripheral tissues and end organs.

This pressure differential is normally maintained by the higher pressure that exists in the arterial system versus the venous system, which drives blood forward. The lower the MAP, the lower this forward flow and the lower the perfusion of the end organs (Fig. 5.5).

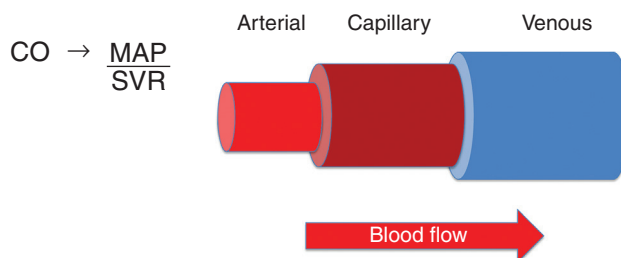


FIG. 5.5 The role of vascular tone in maintaining the vascular pressure head. *CO*, Cardiac output; *MAP*, mean arterial pressure; *SVR*, systemic vascular resistance

Think of a garden hose that is losing water flow, not allowing you to reach a bush that you are watering. You can put your thumb over the top, which increases the pressure differential and allows you to now reach that bush. However, if the water flow continues to decrease, further occluding the hose with your thumb may allow you to reach the bush, but the overall amount of water that is reaching the bush decreases (Fig. 5.6).

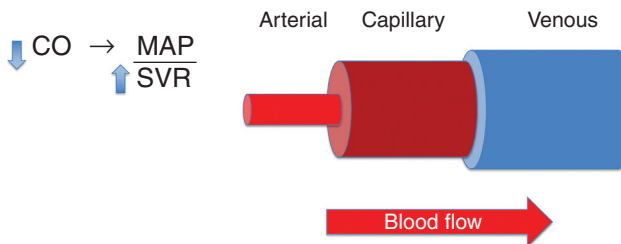


FIG. 5.6 The effect of maintaining mean arterial pressure through vasopressors on forward blood flow. *CO*, Cardiac output; *MAP*, mean arterial pressure; *SVR*, systemic vascular resistance

The same phenomenon occurs when pressors are applied in the setting of decreasing cardiac output. They can be used to a point to maintain this vascular pressure differential, but eventually, the effect is that the overall delivery of blood/oxygen is decreased. This is why you can maintain an adequate MAP with escalating pressors but still experience worsening markers of shock (decreased urine output, lactic acidosis, elevated liver function tests).

The effect is a dose-related response of escalating pressors in shock – with worsening clinical effects, diminishing returns, and increasing toxicity, leading to higher mortality with escalating pressor requirements/dosages. By the time you are reaching for your third or fourth pressor, the expected mortality of your patient has risen precipitously (Fig. 5.7).

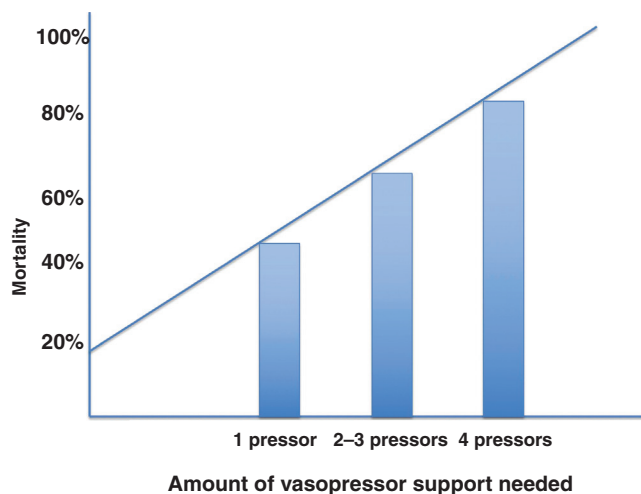


FIG. 5.7 Mortality associated with vasopressor use in cardiogenic shock

This is not to just say that patients who require more pressors are also those more likely to die. It is also saying that when pressors reach escalating dosages, there is less of a chance of them achieving adequate perfusion, and may actually make things worse, leading to the downward spiral of shock.

HOW DO I PREVENT THE DOWNWARD SPIRAL OF SHOCK?

This downward spiraling with interventions making things worse rather than better is a disturbing proposition. Like most downward spirals, the cause comes from not knowing where the spiral is heading and how to counteract. Therefore, let's propose the following stepwise approach, which is designed for assessment of the clinical situation, observation of the effects, and realignment if necessary:

1. Have a diagnosis of shock (obstructive, cardiogenic, distributive, hypovolemic).
2. Implicate a cause of this type of shock (vasodilatory shock due to sepsis due to pneumonia v. obstructive shock due to PE as examples).
3. Address the shock diagnosis with a targeted intervention.
4. Observe the effect of that intervention. Did it improve some parameter or worsen it?

If by step 4 you are not feeling more reassured, then start over and reassess what is going on with the patient.

How Does This Downward Spiral Apply to the Other Components of the Oxygen Delivery Equation?

Returning again to our oxygen delivery equation:

$$\text{DO}_2 \rightarrow \text{CO} \times \text{Hb} \times \text{SaO}_2$$

In a similar way that we talked about how attempts to optimize cardiac output can lead to toxicity, let's now review how optimizing SaO_2 can lead to toxicity.

Like we discussed in [Chapter 4](#), oxygen is able to enter the body efficiently through the alveolar-blood interface and the ability of O_2 to diffuse down its concentration gradient. However, as the

degree of shunt progresses (whether due to worsening pulmonary edema, pneumonia, atelectasis, mucous plugging, etc.), this diffusion gradient becomes not only progressively difficult to maintain but also cannot be improved by the amount of oxygen administered, as illustrated in Fig. 5.8.

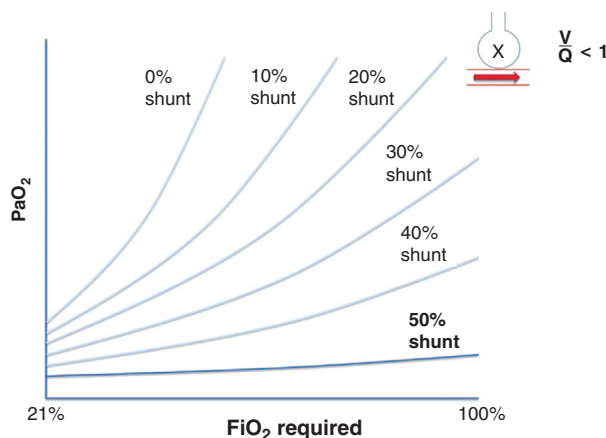


FIG. 5.8 Progressive shunt leading to lower oxygen diffusion despite escalating oxygen administration

At this point, the way to improve oxygenation is to allow for better recruitment of alveoli through augmentation of mean airway pressure.

So to review, the two ways to improve the amount of oxygen that diffuses into the blood from the lungs are to increase the amount of oxygen and to augment the mean airway pressure. Now let's unpack these two strategies, this time with extra attention on the adverse effect of doing so.

WHY IS INCREASING THE AMOUNT OF OXYGEN HARMFUL?

It seems like one of the simplest interventions we can do – if a patient has low oxygen saturations, we increase the amount of oxygen they are receiving. However, oxygen itself can be toxic, especially as concentrations increase. Besides the harmful effects of oxygen administration at the cellular level due to increased free radicals there is a harmful effect of increased oxygen concentrations at the blood-alveolar interface that is worth exploring.

As oxygen concentration increases, this increases the relative concentration of O_2 molecules in the alveolus relative to other gases, mainly nitrogen (N_2). Because oxygen exists at a higher concentration in the alveolus relative to the blood, it always diffuses down its concentration gradient (Fig. 5.9).

N_2 does not have this gradient and therefore does not diffuse. Thus, the higher the concentration of O_2 , the more likely for all of the contents of the alveolus to diffuse into the blood and cause collapse of the alveolar unit (Fig. 5.10).

HARMFUL EFFECTS OF RAISING MEAN AIRWAY PRESSURE

Remember that as shunt increased, the way to improve overall oxygenation becomes dependent on the ability to recruit additional alveoli, by raising the mean airway pressure, mainly through increasing positive end-expiratory pressure (PEEP), as there is more time spent in expiration than in inspiration as illustrated in Fig. 5.11.

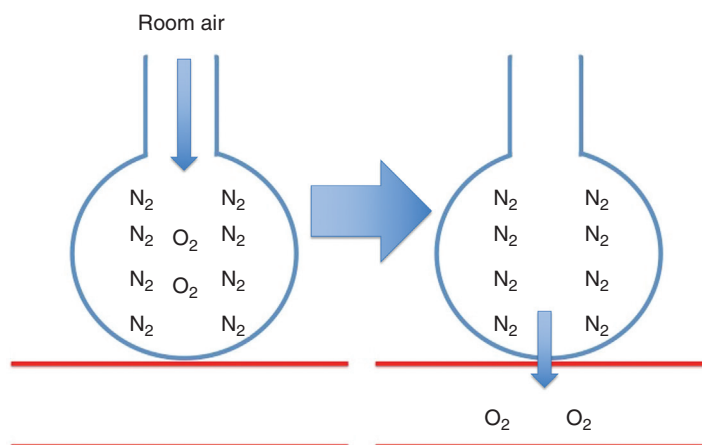


FIG. 5.9 The effect of nitrogen in maintaining alveolar patency

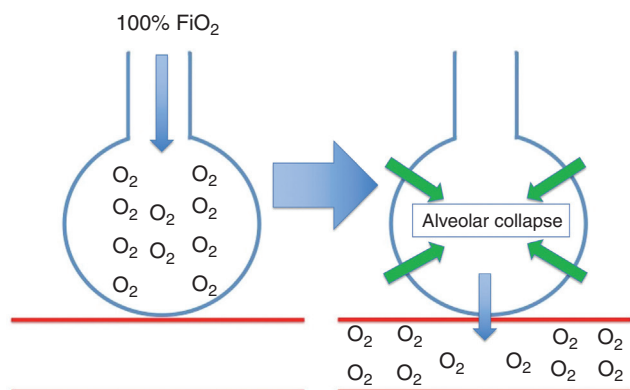


FIG. 5.10 High oxygen requirements leading to alveolar collapse

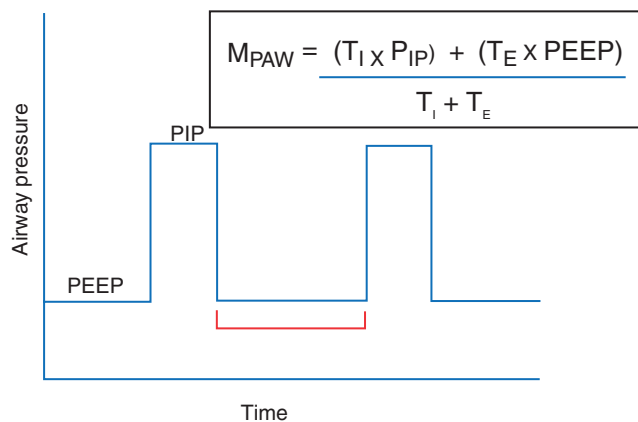


FIG. 5.11 The effect of PEEP on mean airway pressure due to longer time of expiration. *PEEP*, Peak end-expiratory pressure; *PIP*, peak inspiratory pressure

However, doing so has the adverse effect of increasing the overall pressure or decreasing the driving pressure. This is exacerbated as the overall pulmonary compliance decreases due to worsening progression of respiratory disease, whether due to edema, infection, inflammation/acute respiratory distress syndrome, plugging, etc. (Fig. 5.12).

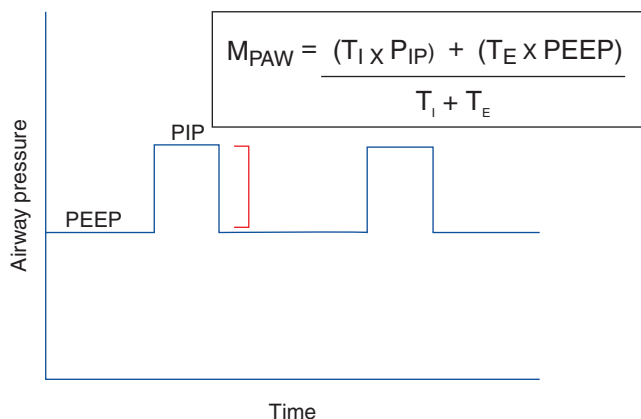


FIG. 5.12 The effect of raising PEEP on overall airway pressures. *PEEP*, Peak end-expiratory pressure; *PIP*, peak inspiratory pressure

As compliance starts to worsen, the overall driving pressure either has to increase or the tidal volume and PEEP has to decrease. The consequence is a worsening of oxygenation and ventilation/ CO_2 removal (due to lower PEEP/tidal volume) or higher ventilator pressures have to be maintained, putting the patient at risk for worsening damage from the ventilator. The worsening of oxygenation/ventilation can exacerbate acidosis and vasodilation, increase right heart afterload, and lead to progressive myocardial depression, all with the downstream effect of decreasing cardiac output and further reduction of oxygen delivery.

WHAT ABOUT HEMOGLOBIN?

What about hemoglobin, our efficient compound that increases the delivery of oxygen so precipitously? Can we use hemoglobin to augment oxygen delivery?



It makes intuitive sense, especially if hemoglobin levels are low. If we are having difficulty delivering oxygen through a depressed cardiac output or inability to maintain saturations, why don't we just put more proverbial trucks on the road (Fig. 5.13)?

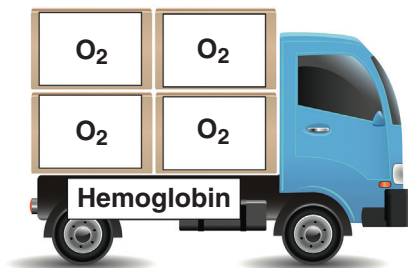


FIG. 5.13 Hemoglobin as a means for augmenting delivery of oxygen.
(From [Bannosuke/Shutterstock.com](https://www.shutterstock.com).)

Unfortunately, like all the interventions that we have explored, hemoglobin augmentation through blood transfusion is subject to both diminishing returns with progressive administration, as well as toxicity that increases with progressive administration.

The diminishing returns center on the limitations of blood transfusion – the reality that transfused blood does not carry and deliver oxygen to the same degree that native blood does.

Additionally, there exist a multitude of toxicities that are well described with blood – including inflammatory effects, volume effects, and association with immune depressions and infections. These can impact organs throughout the body.

THE FINAL POINT...

Hopefully this chapter endows you with a sense of the dose-related response of critical care interventions designed to augment the delivery of oxygen. To be able to properly titrate the amount of support needed, we need to first understand the physiology of what is going on with the patient, what is deficient, and how we can support that deficiency in a way that is as effective and associated with as little toxicity as possible.

Realizing the toxicity of progressive up-titration of supportive care is an essential part of understanding the role of extracorporeal support. This is because, as we will now uncover, it is understanding the role of extracorporeal support in sparing this toxicity that will form the basis of how we approach extracorporeal membrane oxygenation (ECMO) and the role of ECMO in the care of critically ill patients.

Let's return now to our initial graph, showing the dose-related response of support with mortality, this time superimposing a proposed relationship that exists for specific patients who are now supported on ECMO, as represented by the green line (Fig. 5.14).

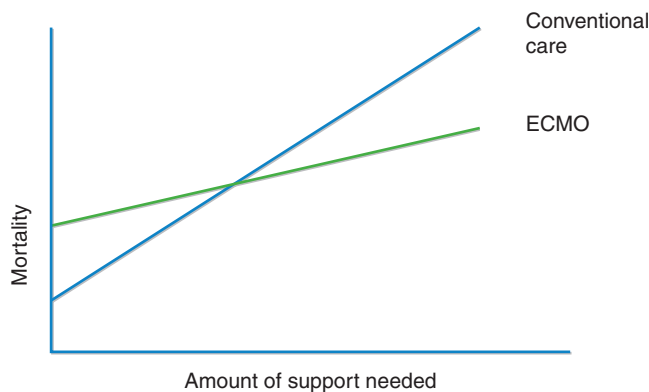


FIG. 5.14 ECMO as a potential option for decreasing the toxicity of high-dose critical care support

What if there is an intervention that we can do that can curb the toxicity of what we are doing to support our patients? What if that intervention involved more risk, cost, and expertise? Would it be worth it? As you can see, this intervention called ECMO, which we will spend the remainder of the book exploring and understanding, can only be useful if it is helping with the fundamental need of allowing for a sparing of toxicity, morbidity, and mortality, in order to offer the possibility of a better outcome for patients.

SUGGESTED READING

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